

Discussion Log

StockVolatilitySight — CPSC 440

Running record of important decisions, divergences from the proposal, and rationale. Keep entries concise and focused — this is a working document for the team, not a plan artifact.

Conventions.

- The proposal (`plan/Research 440 Proposal.md`) frames the research question and is the source of truth for *intent*.
 - The plan doc (`plan/hmm-lstm_volatility_forecast_11d292d8.plan.md`) is a hypothesis / implementation log. Empirical results can and do diverge from what the plan predicted — that’s the research finding, not a bug. Log divergences here with neutral framing.
 - When implementation diverges from the *proposal*, log it here with a rationale.
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Current State (updated 2026-04-23, post-rerun)

Pipeline status. Phases 1–6 of the plan are complete. Full pipeline re-run from a fresh clone today (nb 01 → 04 → 05 → `src/ensemble.py` → nb 06). HMM selected (GMMHMM, 2 states, 2 mixture components, full covariance). All three LSTMs trained. Ensemble evaluated on 2020-06-30 → 2026-03-16 ($n = 1,434$).

Headline empirical result (fresh run, notebooks/06_ensemble_eval.ipynb):

Model	MSE	MAPE	Prior-run MSE
Naive (<code>rolling_std_21</code>)	2.2e-5	34.6%	2.2e-5
Baseline LSTM	1.6e-5	25.1%	1.5e-5
Calm LSTM	4.1e-5	34.9%	2.0e-5
Volatile LSTM	8.5e-5	119.4%	8.5e-5
Ensemble	2.6e-5	31.9%	1.7e-5

Diebold–Mariano significance tests (nb 06 §6, first time actually executed):

Comparison	Loss	DM	p	Verdict ($\alpha = .05$)
Naive vs Baseline (full)	MSE	+1.36	0.173	tie
Naive vs Baseline (full)	MAE	+2.46	0.014	Baseline wins
Baseline vs Ensemble (full)	MSE	-1.75	0.080	tie (marginal, $\alpha = .10$)
Baseline vs Ensemble (full)	MAE	-2.00	0.046	Baseline wins
Baseline vs Ensemble (volatile, $n = 27$)	MSE	+0.62	0.542	tie
Baseline vs Ensemble (volatile, $n = 27$)	MAE	+0.22	0.831	tie

On aggregate the baseline LSTM beats the HMM-LSTM ensemble — significantly on MAE ($p = 0.046$), marginally on MSE ($p = 0.080$). On the 27 strictly-volatile test days the ensemble is *directionally* better (MSE 5e-5 vs 8e-5) but the sample is too small to reject ($p > 0.5$). Regime separation helps where it should; the calm regime dominates aggregate metrics.

Training stochasticity concern. This run’s ensemble MSE is 2.6e-5 vs the 1.7e-5 in the previously-saved notebook output — a $\sim 50\%$ difference from the same config, with different Optuna/PyTorch numerical noise. The calm LSTM doubled its MSE (4.1e-5 vs 2.0e-5). **Single-seed results are not reliable for ranking claims** — before locking in any narrative we should run k seeds and report mean $\pm$ std.

Known reproducibility gaps (four now).

- `data/processed/train|val|test.parquet` predate the Phase-6 addition of `relative_volume_21d`. Cloning and running nb 04 directly $\rightarrow$ `KeyError`. Fix: re-run nb 01 (which calls `features.build_features` and saves with the column).
- `src/ensemble.py` is an undocumented step between nb 05 and nb 06. Nb 06 cell 4 reads `data/processed/test_predictions.parquet`, which `ensemble.py` produces. Neither the README nor the notebooks call this out. The nb 05 $\rightarrow$ nb 06 chain fails with `FileNotFoundError` without this step. Discovered today when running end-to-end.
- `data/processed/test_predictions.parquet` is not tracked (produced by the step above).
- LSTM checkpoints (`models/*.pt`) are gitignored.

Open items to decide.

1. Is the aggregate-loss / volatile-day-win tradeoff acceptable for the writeup, or do we want to revisit gating? (Proposal explicitly listed a learned gating network as out of scope.)
2. Volatile LSTM outputs ~ 0.0175 nearly constantly and pulls the ensemble up on calm days — does this need a fix (clipping, higher gating threshold) or is it part of the finding?
3. `config.VAL_END = "2020-04-30"` means val now includes COVID. The plan text still says “2016–2019.” Decide whether to update the plan text or realign the config.
4. **Multi-seed robustness study.** Re-run LSTM training under $k \geq 5$ random seeds to get mean $\pm$ std of the metrics. The current single-run swing from ensemble-MSE 1.7e-5 to 2.6e-5 means any “baseline beats ensemble” claim is fragile.
5. Document the nb 05 $\rightarrow$ `ensemble.py` $\rightarrow$ nb 06 step in README.md (or bundle `ensemble.py` invocation into nb 06’s setup cell).

2026-04-23 — Plan summary and proposal/plan discrepancies

Plan summary

Regime-aware hybrid model for 21-day forward realized volatility on SPY:

1. **Data + features:** OHLCV + put/call + AAI sentiment $\rightarrow$ log returns, abs returns, ranges, rolling stats; chronological split; normality diagnostics to justify HMM choice.
2. **HMM regime detection:** Gaussian vs GMM-HMM, 1st- vs 2nd-order (via augmented state space), BIC/LL comparison, Viterbi sanity check against 2008/2020/2022 crises.
3. **LSTMs:** baseline (full data) + calm + volatile, PyTorch CPU, log-target with MSE, Optuna TPE tuning.
4. **Ensemble:** $y = P(\text{calm} \mid O) \cdot y_{\text{calm}} + P(\text{volatile} \mid O) \cdot y_{\text{volatile}}$.
5. **Evaluation:** MSE/RMSE/MAE vs baseline LSTM on 2020-present.

Plan-era result (pre-Phase-6): ensemble MSE 0.00167 vs baseline 0.00246 on the 2020-present holdout — the regime-split hypothesis was initially validated. See “Current State” above for the post-Phase-6 empirical update.

Discrepancies between proposal and plan

#	Proposal says	Plan / code does	Rationale
1	Targets SPY, XEQT, and MAG7 individual stocks	SPY only	XEQT has insufficient history; MAG7 dropped from scope. Documented in plan.
2	Train 2005–2015, Val 2015–2020, Test remaining	Train 2004–2015, Val 2016–2019, Test 2020-present	Not explicitly rationalized in plan (git log shows a revert on the split). Non-overlapping boundaries are cleaner but the drift from the proposal is undocumented. Open item.
3	Inputs include raw OHLCV	Phase 6 removed raw OHLCV; kept log_return, abs_return, oc_return, intraday_range, relative_volume_21d	Raw prices caused non-stationary drift in predictions; stationarity fix dropped MAPE from ~600% to ~25%. Tried, didn't work — legitimate divergence.
4	Compare Gaussian vs GMM-HMM; explore higher-order Markov “if time permits”	Both done (GMM-HMM and 2nd-order via augmented state space)	No discrepancy.
5	Reference baselines: baseline LSTM + “considering” a simple tree/NN	Baseline LSTM + naïve rolling_std_21 (no tree model)	Plan picked a different simple baseline. Proposal's tree regressor was never implemented.
6	Regime-specific LSTMs train on regime-partitioned data	Volatile-LSTM val fallback: 0 volatile-dominant windows in 2016–2019, so early-stopping uses the unfiltered val set	Practical necessity — the 2016–2019 val window is almost entirely calm. Methodological caveat: volatile LSTM's model selection isn't purely regime-matched.
7	Learned gating network — considered, out of scope	Out of scope	No discrepancy.
8	Non-price inputs: put/call volume (Ho et al., 2012) and AAI weekly sentiment (Gupta et al., 2023) as key features	Scaffolded but inert. SENTIMENT_FEATURES + PUTCALL_FEATURES defined in config.py since commit 888a861 (2026-04-16), loaders exist in src/data_loader.py, but the CSVs were never downloaded to data/raw/ and neither group is referenced by HMM_FEATURES or LSTM_BASELINE_FEATURES. Verified via git log: never added to the active feature lists, never removed — just never activated.	HMM + LSTMs train on price/volume only. Significant divergence from the proposal's core “non-price inputs matter for vol” thesis. Options for the paper: (a) download the CSVs and retrain to restore proposal alignment; (b) explicitly note the scope-down in methods + limitations.

Most consequential to flag in the writeup: #1 (scope narrowing to SPY), #3 (stationarity-driven feature change), #6 (volatile-LSTM val fallback), **#8 (sentiment + put/call never wired in — materially weakens the “non-price signals matter” thesis)**. #2 is the open item — worth deciding whether to retro-justify or realign.

2026-04-23 — Repo survey: code, notebooks, data sanity

Data we have

Checked-in processed data (data/processed/):

- `features_all.parquet` — full feature table, 5,584 rows, 2004-01-05 → 2026-03-16.
- `train.parquet` (3,020 rows, 2004-01-05 → 2015-12-31), `val.parquet` (1,089 rows, 2016-01-04 → 2020-04-30), `test.parquet` (1,475 rows, 2020-05-01 → 2026-03-16).
- `regime_probabilities.parquet` — HMM soft probs + Viterbi labels, 5,564 rows. Overall regime mix: 96.3% calm, 3.7% volatile (matches plan).

Columns present in train/val/test (20): `Open`, `High`, `Low`, `Close`, `Volume`, `log_return`, `abs_return`, `oc_return`, `intraday_range`, `log_volume`, `rolling_mean_{5,10,21}`, `rolling_std_{5,10,21}`, `rolling_abs_mean_{5,10,21}`, `realized_vol_21d`.

Sanity spot-checks on train pass: `log_return` mean ≈ 0.0004 , std ≈ 0.012 (✓ daily SPY scale); `log_volume` $\in [16.5, 20.6]$ (✓ log of millions); target `realized_vol_21d` positive, same scale as `rolling_std_21`; no NaNs after rolling warmup. Target formula in `features.py:154` correctly implements the proposal: forward-looking, demeaned, window = 21 —

$$\sigma_t = \sqrt{\frac{1}{m} \sum_{j=1}^m (r_{t+j} - \bar{r})^2}, \quad m = 21.$$

Code layout

`src/` holds all modules (not `models/`, which only holds serialized `.joblib` artifacts). Key files:

- `features.py` — engineers all features and target; matches proposal math.
- `hmm_model.py` — Gaussian vs GMM-HMM comparison, penalised EM with restarts, 1st- vs 2nd-order via state-space augmentation.
- `lstm_model.py` — stacked LSTM + linear head, sliding-window `Dataset`, log-space prediction.
- `train_LSTM_baseline.py`, `train_LSTM_regime.py` — Optuna TPE tuning → final retrain.
- `ensemble.py` — weighted-sum ensemble keyed on HMM soft probs.
- `config.py` — single source of truth for features, splits, HMM/LSTM search spaces.

HMM artifacts in `models/`: `hmm_winner.joblib` = `GMMHMM(states=2, n_mix=2, cov=full)`, val LL = 9057.28, BIC = -15925.74. Volatile state is state 1. `HMM_DURATION_PENALTY=3000` (approximately the number of training observations) prevents degenerate rapid-switching during EM restarts.

Current test-set results

From `notebooks/06_ensemble_eval.ipynb`, 2020-06-30 $\rightarrow$ 2026-03-16, $n = 1,434$.

Model	MSE	RMSE	MAE	MAPE
Naive (<code>rolling_std_21</code>)	0.000022	0.00470	0.00313	34.6%
Baseline LSTM	0.000015	0.00390	0.00246	25.3%
Calm LSTM	0.000020	0.00442	0.00279	29.5%
Volatile LSTM	0.000085	0.00922	0.00857	119.4%
Ensemble	0.000017	0.00415	0.00271	28.8%

Regime-stratified (HMM confidence > 0.8): on strictly-calm days ($n = 1,407$) baseline wins (MSE $1e-5$ vs $2e-5$); on strictly-volatile days ($n = 27$) **ensemble wins** (MSE $5e-5$ vs $1e-4$).

Findings worth flagging

1. **Empirical ordering differs from the plan’s hypothesis.** The plan doc anticipated ensemble $>$ baseline (MSE 0.00167 vs 0.00246, pre-Phase-6). Current post-Phase-6 notebook-06 output shows baseline ($1.5e-5$) slightly ahead of ensemble ($1.7e-5$) on aggregate — but ensemble beats baseline on strictly-volatile days ($n = 27$, MSE $5e-5$ vs $1e-4$). Not a failure: the plan is a hypothesis doc, and this *is* the research result. Worth framing clearly in the writeup as “regime separation helps on volatile regimes specifically; the calm regime dominates 2020–2026 so the baseline is hard to beat on aggregate.”
2. **Volatile LSTM is overpredicting on calm days.** Its test MAPE is 119% — when the HMM is wrong about the regime (or uncertain), the volatile model’s near-constant high output (~ 0.0175) pulls the ensemble up. Options: (a) clip volatile predictions when `p_volatile` is small, (b) use a higher gating threshold, (c) revisit the learned-gating-network idea that the proposal put out of scope.
3. **Val split now goes to 2020-04-30, not 2019-12-31.** `config.VAL_END = "2020-04-30"` — val *includes* the COVID crash. The plan text and plan table both say “2016–2019” val. This change means the volatile-LSTM val-fallback issue (plan Phase 3b) may no longer apply, since COVID gives the val set ~ 2 months of volatile-dominant windows. **Discrepancy vs plan and vs the #2 item in the earlier discrepancy table.** The config comment on line 22 is also wrong (says test is “2020-01-01 $\rightarrow$ present” but test actually starts 2020-05-01).
4. **Saved parquets are stale vs the code.** `config.LSTM_BASELINE_FEATURES` requires `relative_volume_21d`, and `features.build_features` computes it (line 298), but the checked-in `train/val/test.parquet` files do not contain the column. Re-running notebook 01 would fix it; trying to retrain LSTMs against the committed data will `KeyError`. Also, `data/processed/test_predictions.parquet` (which notebook 06 loads in cell 4) is not in the tree — results only reproducible from a full local regeneration.
5. **LSTM checkpoints are intentionally not tracked.** `.gitignore` excludes `models/*.pt`, `models/*.pth`. This is a choice (size), but combined with (4) it means a fresh clone can’t reproduce the ensemble without re-running notebooks 01, 03, 04, 05, 06 in order. Worth a note in the README if we care about reproducibility for the graders.
6. **Raw OHLCV still lives in the parquets.** Phase 6 removed them from `LSTM_BASELINE_FEATURES`, but the columns remain in the DataFrames (just unused). Not a bug — feature selection happens

at the config level — but worth knowing if anyone’s debugging feature plumbing.

2026-04-23 (later) — Added Diebold–Mariano forecast significance testing

What prompted this

In discussion of the metrics table, two methodological concerns surfaced:

1. **Raw MSE is scale-misleading at this target scale.** Realized volatility sits in the 0.005–0.05 range, so MSE is necessarily tiny. Reporting “Baseline beats Naive by 32% MSE” ($2.2\text{e-}5 \rightarrow 1.5\text{e-}5$) made the gap sound dramatic, but the absolute numbers convey little about practical significance. MAPE ($34.6\% \rightarrow 25.3\%$, a 9.3pp absolute reduction in relative error) and RMSE ($0.00470 \rightarrow 0.00390$, $\sim 17\%$ lower) are more interpretable — each forecast is off by roughly a quarter of the true vol, not a 30th of anything.
2. **All rankings were point estimates.** We had not formally tested equal predictive accuracy, so we couldn’t distinguish “genuinely more accurate” from “slightly lower empirical loss on this sample.” This matters most for the headline **Baseline vs Ensemble** comparison, where the RMSE gap is only $\sim 6\%$ (0.00390 vs 0.00415) — well within the range where stochastic variation could flip the ordering.

The second point is a legitimate weakness in the current writeup: we were making ranking claims without showing they were significant.

What was added

notebooks/06_ensemble_eval.ipynb — new **Section 6** (cells 14–17):

- `diebold_mariano()` helper implementing the Diebold–Mariano (1995) test with the Harvey–Leybourne–Newbold (1997) small-sample correction. Newey–West HAC variance with Bartlett kernel, $\text{lag} = h - 1 = 20$ to account for autocorrelation from our overlapping 21-day targets. Supports MSE, MAE, and MAPE loss.
- Three comparisons, each run with both MSE and MAE loss for robustness:
 1. **Naive vs Baseline LSTM** — sanity floor.
 2. **Baseline vs Ensemble** (full test, $n = 1,434$) — primary research question.
 3. **Baseline vs Ensemble** on strictly-volatile days ($p_{\text{volatile}} > 0.8$, $n = 27$) — the regime where the ensemble is supposed to help.
- A formatted results table with DM statistic, p -value, significance code, and $\alpha = 0.05$ verdict.
- Interpretation notes clarifying that the strictly-volatile comparison is severely underpowered and should be read as directional evidence only.

Status: pending execution

`data/processed/test_predictions.parquet` and the `models/*.pt` LSTM checkpoints are not tracked (see “Known reproducibility gaps”), so the DM cells cannot run on the current fresh checkout.

The code is in place and will execute on the next full regeneration of notebooks 01 → 06.

Priors — what we expect when it runs

Test	Prior	Reasoning
Naive vs Baseline (full)	Strongly reject ($p < 0.01$)	RMSE gap ~17% with $n = 1,434$ — ample power
Baseline vs Ensemble (full)	Ambiguous	Gap only ~6% RMSE; plausibly within noise
Baseline vs Ensemble (volatile)	Likely fail to reject	$n = 27$ is too small to reject despite large point-estimate gap (ensemble halves MSE)

Why this matters for the paper

DM results let us calibrate our claims:

- If **naive vs baseline** rejects → “the deep model learns structure beyond persistence.” ✓ clean claim.
- If **baseline vs ensemble (full)** does *not* reject → reframe headline from “baseline slightly beats ensemble” to “**baseline and ensemble are statistically tied on aggregate, and ensemble is directionally better on volatile-dominant days.**” This is both more honest and a stronger-feeling paper than claiming the ensemble lost.
- If **baseline vs ensemble (volatile)** does *not* reject → we cannot make a significance claim on the volatile subset. Report as directional evidence plus an explicit note on low power, and suggest larger/longer test windows (extend the test period before 2020 or include additional crisis periods) as future work.

Weakness surfaced (for the paper’s limitations section)

Until this point we were reporting model rankings without testing them. Identified during review; now remedied for the aggregate and volatile comparisons. A remaining weakness is test-set power for regime-conditional claims: the 2020-06-30 → 2026-03-16 window contains only 27 days meeting `p_volatile` > 0.8, which is insufficient to detect anything short of huge effects. This is a structural limitation of using SPY (calm-dominated) on a post-COVID test window rather than a flaw in the model.

2026-04-23 (evening) — Pipeline re-run from fresh clone; DM results executed

Motivation

Imported the repo fresh and attempted to reproduce the pipeline end-to-end. Two things we wanted: (1) confirm the reproducibility gaps we’d identified were real, (2) actually produce `test_predictions.parquet` so the Diebold–Mariano cells added earlier today could run.

Reproducibility issues found (in the order we hit them)

1. **relative_volume_21d missing from committed parquets** — `train_LSTM_baseline.py` failed immediately with `KeyError: "Columns missing from dataframe: ['relative_volume_21d']"`. Fix: re-run nb 01. Verified this is a pre-existing gap (git status confirmed we hadn't touched `config.py`, `src/`, or `data/processed/`). Same failure any teammate would hit from a fresh clone.
2. **src/ensemble.py step is undocumented** — after nb 04 + nb 05 produced the three LSTM .pt files, nb 06 failed with `FileNotFoundError: data/processed/test_predictions.parquet`. This file is produced by running `python src/ensemble.py` as an intermediate step. The README lists only notebooks; neither nb 05 nor nb 06 mentions that an external script must run between them. Fix: explicitly document the step in README, or bundle the invocation into nb 06's setup cell.
3. **Package install was ad-hoc**. I installed `optuna`, `seaborn`, `yfinance` piecemeal as errors hit, instead of `pip install -r requirements.txt` up front. Lesson for the log: start with the README's setup block.

Fresh-run metrics ($n = 1,434$, test window 2020-06-30 → 2026-03-16)

Model	MSE	RMSE	MAE	MAPE
Naive (rolling_std_21)	0.000022	0.00470	0.00313	34.6%
Baseline LSTM	0.000016	0.00398	0.00257	25.1%
Calm LSTM	0.000041	0.00638	0.00343	34.9%
Volatile LSTM	0.000085	0.00922	0.00857	119.4%
Ensemble	0.000026	0.00511	0.00310	31.9%

Regime-stratified ($p > 0.8$): strictly-calm $n = 1,407$ — Baseline MSE 0.00001, Ensemble MSE 0.00003 (baseline wins). Strictly-volatile $n = 27$ — Baseline MSE 0.00008, Ensemble MSE 0.00005 (ensemble wins directionally).

Diebold–Mariano results

Sign convention: positive DM statistic means the first-listed model has higher loss (so the second-listed model is more accurate). Newey–West HAC with Bartlett kernel, lag 20; HLN small-sample correction; two-sided t -test with $df = n - 1$.

Comparison	Scope	Loss	DM	p	Verdict ($\alpha = .05$)
Naive vs Baseline	full $n = 1,434$	MSE	+1.36	0.173	tie
Naive vs Baseline	full $n = 1,434$	MAE	+2.46	0.014	Baseline wins
Baseline vs Ensemble	full $n = 1,434$	MSE	−1.75	0.080	tie (marginal)
Baseline vs Ensemble	full $n = 1,434$	MAE	−2.00	0.046	Baseline wins
Baseline vs Ensemble	volatile $n = 27$	MSE	+0.62	0.542	tie
Baseline vs Ensemble	volatile $n = 27$	MAE	+0.22	0.831	tie

Priors vs reality

Test	Prior (pre-run)	Actual	Assessment
Naive vs Baseline	Strongly reject ($p < 0.01$)	MSE $p = 0.17$ (tie), MAE $p = 0.014$ (reject)	Weaker than predicted on MSE. The squared-error loss has higher variance, so the HAC-corrected test has less power. MAE gives the clean claim.
Baseline vs Ensemble (full)	Ambiguous	MSE $p = 0.08$, MAE $p = 0.046$	Prior confirmed. Result is borderline — depending on loss and α , baseline either marginally wins or ties.
Baseline vs Ensemble (volatile)	Likely fail to reject despite large gap	$p = 0.54 / 0.83$	Prior confirmed. $n = 27$ is simply too small.

Defensible claims for the paper

- “The baseline LSTM is more accurate than the 21-day rolling-volatility persistence baseline (DM MAE test, $p = 0.014$).” $\leftarrow$ strongest claim; use MAE not MSE.
- “The baseline LSTM matches or slightly outperforms the HMM-LSTM ensemble on aggregate (DM, $p = 0.046$ MAE, $p = 0.080$ MSE).” $\leftarrow$ honest framing. Not “the ensemble fails.”
- “On volatile-dominant subperiods, the ensemble is *directionally* more accurate (point-estimate MSE 0.00005 vs 0.00008), but the $n = 27$ subsample is underpowered to reject equal accuracy at $\alpha = 0.05$.” $\leftarrow$ directional with explicit caveat.

Training stochasticity — a new concern

Compared to the previously-saved nb 06 output (a prior run by a teammate):

Model	Prior saved MSE	This-run MSE	Change
Baseline	1.5e-5	1.6e-5	+7%
Calm	2.0e-5	4.1e-5	+ 105%
Volatile	8.5e-5	8.5e-5	0%
Ensemble	1.7e-5	2.6e-5	+ 53%

Same config, same seeds in `config.py` (RANDOM_STATE=42, LSTM_RANDOM_STATE=42), same data. The gap comes from (a) Optuna’s TPE exploration — different trial sequences can converge to different hyperparameter neighbourhoods, (b) CPU PyTorch non-determinism without `torch.use_deterministic_algorithms`.

Implication: ranking claims based on a single training run are fragile. The baseline-vs-ensemble MSE gap went from 13% (prior) to 63% (this run). Before locking in the narrative for the paper,

we should run $k \geq 5$ seeds and report mean $\pm$ std on each metric, or at minimum run 3 seeds and report the range. This is a real limitation to cite in the paper’s methodology/limitations section.

Why this entry matters for the paper writeup

- Shows the empirical result is genuinely nuanced, not a clean win or loss.
 - Documents that we identified and tested methodological weaknesses (no significance testing $\rightarrow$ added DM; single-run results $\rightarrow$ flagged multi-seed need).
 - Provides a transparent trail: priors $\rightarrow$ actual results $\rightarrow$ assessment, which is the kind of thing reviewers value.
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2026-04-24 — HAR-RV benchmark design: target-formula alignment

Why this entry exists

Before coding `src/har_rv.py` we pin down exactly what target HAR-RV will predict and exactly what predictors it will use. This is not a neutral choice — Corsi’s original HAR-RV (2002/2004) uses a different target and a different predictor convention than our project’s `realized_vol_21d`. Getting this wrong would make the benchmark non-comparable to our LSTMs / ensembles.

User reviewed `ignore/ssrn-626064.pdf` (Corsi’s original HAR-RV paper) and asked:

1. Is HAR-RV predicting backward volatility? Is that cheating?
2. Is $\sqrt{\text{variance}}$ = volatility as assumed in our target?
3. Document precisely where our target, Corsi’s HAR-RV, and our adjusted HAR-RV agree and disagree.

This entry answers all three.

Three formulas, side by side

1. Our target (what every LSTM / ensemble in this project predicts). Defined in `src/features.py::add_realized_volatility()` (lines 167–216). Andersen & Bollerslev (1998) form, adapted for daily data with demeaning:

$$\sigma_t = \sqrt{\frac{1}{m} \sum_{j=1}^m (r_{t+j} - \bar{r})^2} \quad (1)$$

with $m = 21$, $\bar{r}$ computed over the *same* forward window $[t+1, t+m]$.

- **Forward-looking:** aggregates returns from $t+1$ to $t+m$.
- **Demeaned:** AB98 subtract the window mean before squaring; appropriate for non-high-frequency data.
- **Normalisation** $1/m$: population-variance convention (not $1/(m-1)$).
- **Units:** volatility (sqrt of variance), same units as returns.

2. Corsi’s original HAR-RV (ssrn-626064.pdf, equation 13).

$$RV_{t+1d}^{(d)} = c + \beta^{(d)} RV_t^{(d)} + \beta^{(w)} RV_t^{(w)} + \beta^{(m)} RV_t^{(m)} + \omega_{t+1d} \quad (2)$$

with the components defined on intraday data (his equation 3):

$$RV_t^{(d)} = \sqrt{\sum_{j=0}^{M-1} r_{t-j\Delta}^2} \quad (\text{un-demeaned; } M \text{ intraday returns}) \quad (3)$$

$$RV_t^{(w)} = \frac{1}{5} (RV_t^{(d)} + \dots + RV_{t-4d}^{(d)}) \quad (\text{simple 5-day average}) \quad (4)$$

$$RV_t^{(m)} = \frac{1}{22} (RV_t^{(d)} + \dots + RV_{t-21d}^{(d)}) \quad (\text{simple 22-day average}) \quad (5)$$

Structural differences from our target:

Aspect	Our target (AB98)	Corsi HAR-RV
Forecast horizon	21 days ahead	1 day ahead
Observation frequency	daily	intraday (M ticks/day)
Demeaning	yes	no (intraday mean ≈ 0)
Normalisation	$1/m$	sum (daily) / simple avg (w,m)
“Monthly” window	21 trading days	22 trading days
Units	volatility ($\sqrt{\cdot}$)	volatility ($\sqrt{\cdot}$)

Crucial agreement: Corsi is in *volatility units*, not variance. His equation (3) is $\sqrt{\cdot}$, and his regression (equation 13) is linear in those sqrt-form quantities. So $\sqrt{\text{variance}} = \text{volatility}$ is consistent between Corsi and our target. This directly answers the user’s question 2.

Not cheating: the regression has future vol on the LHS and past-vol aggregates on the RHS. Same structure as our LSTMs — past features $\rightarrow$ future target — just OLS on three hand-engineered backward summaries instead of a deep model.

3. Our adjusted HAR-RV (what src/har_rv.py will implement).

$$\underbrace{\text{realized_vol_21d}(t)}_{y = \text{OUR EXACT TARGET}} = c + \beta_d \cdot RV_d(t) + \beta_w \cdot RV_w(t) + \beta_m \cdot RV_m(t) + \varepsilon \quad (6)$$

with predictor components at time t , using past returns only:

$$RV_d(t) = |r_t| \quad (\text{un-demeaned; mandatory, see below}) \quad (7)$$

$$RV_w(t) = \sqrt{\frac{1}{5} \sum_{j=t-4}^t (r_j - \bar{r}_5)^2} \quad (\text{demeaned; matches target formula}) \quad (8)$$

$$RV_m(t) = \sqrt{\frac{1}{21} \sum_{j=t-20}^t (r_j - \bar{r}_{21})^2} \quad (\text{demeaned; matches target formula}) \quad (9)$$

Key invariants:

- **y unchanged:** same `realized_vol_21d` column that every other predictor in this project (A, H, B, O, naive) is evaluated against. MSE / MAE / MAPE / Diebold–Mariano all directly comparable.
- **5- and 21-day components demeaned:** structurally identical to the target’s formula at shorter backward windows.
- **1-day component un-demeaned:** forced by mathematics. Demeaning a single observation is degenerate: $\text{mean}([r_t]) = r_t$, so $(r_t - \bar{r})^2 = 0$. Corsi’s convention handles this the same way.
- **21-day monthly window (not 22):** aligns with `config.ROLLING_WINDOWS` and the target horizon. One-day cosmetic difference from Corsi.

Why the hybrid instead of pure Corsi or pure “match-target-everywhere”

Two pure options were considered and rejected:

- **Pure Corsi** (all predictors un-demeaned, 22-day monthly): literature-faithful but gratuitously inconsistent with our target’s demeaned formula, for windows where demeaning is well-defined and numerically matters.
- **Pure “match target everywhere”** (all three predictors demeaned at 1/5/21): the 1-day component mathematically collapses to 0. Benchmark reduces to a 2-predictor regression.

The hybrid preserves: (a) y exactly equal to our target (non-negotiable), (b) predictors sharing the target’s demeaned formula wherever defined (5- and 21-day), (c) the 1-day component in the only sensible form available ($|r_t|$).

HAR-RV is a horizon-parameterised family (resolves “doesn’t HAR-RV predict tomorrow?”)

A reasonable follow-up: Corsi’s equation 13 has $RV_{t+1d}^{(d)}$ on the LHS — i.e., *tomorrow’s* vol. How does that apply to our 21-day-forward problem?

“HAR-RV” in the literature is a family parameterised by the forecast horizon h . The RHS (backward 1/5/22-day vol components) never changes; only the LHS target changes:

$$h = 1 \quad \rightarrow \quad RV_{t+1d} \quad (\text{Corsi’s original, 1-day ahead}) \quad (10)$$

$$h = 5 \quad \rightarrow \quad RV_{t+1:t+5} \quad (\text{weekly forecast}) \quad (11)$$

$$h = 22 \quad \rightarrow \quad RV_{t+1:t+22} \quad (\text{monthly forecast — our case, } h \approx 21) \quad (12)$$

Andersen–Bollerslev–Diebold (2007), Corsi–Renò (2012), Bollerslev–Patton–Quaedvlieg (2016), and Christensen–Siggaard–Veliyev (2023) all routinely report HAR-RV at $h = 1, 5, 22$ side-by-side and all call it “HAR-RV”. The $h \approx 21$ variant is the standard monthly-horizon benchmark. Our adaptation is not exotic — it’s HAR-RV at the horizon matching our target.

Practical expectation at $h = 21$: the monthly predictor $RV_m(t)$ will dominate the regression (large β_m), because a backward 21-day vol window is structurally the closest predictor to a forward 21-day vol target. This is itself an empirical finding worth reporting — it’s essentially why naive `rolling_std_21` (which is exactly the numerator of RV_m) is already a competitive baseline.

Role of the LHS: target during training vs forecast during prediction

Second follow-up: when we write $\text{realized_vol_21d}(t) = c + \beta_d \cdot RV_d(t) + \dots + \varepsilon$, is the LHS the *actual* value or the *predicted* value?

Answer: **both, depending on phase**. This is the standard regression-ML setup but worth spelling out since HAR-RV and our LSTMs share it.

Phase	LHS is ...	Computed how
Training (fit OLS)	observed ground truth $y_\tau = \text{realized_vol_21d}(\tau)$ for historical $\tau \leq \text{train_end}$	from returns $r_{\tau+1}, \dots, r_{\tau+21}$ that have already been observed (everything in training is past, so both X and y are knowable)
Prediction (at test date t)	forecast $\hat{y}_t$	plug $X_t = [RV_d(t), RV_w(t), RV_m(t)]$ into the fitted equation; the true y_t isn't observable until 21 trading days have elapsed
Evaluation (on test rows)	both available	compare $\hat{y}_t$ (forecast) to y_t (realized, now computable in hindsight); MSE / MAE / DM tests run on the resulting $(\hat{y}_t, y_t)$ pairs

Why this is not circular: temporal separation. X_t uses only returns at times $\leq t$ (past); $y_t = \text{realized_vol_21d}(t)$ uses returns at times $t+1, \dots, t+21$ (future). At time t , X_t is available but y_t is not. The model's job is to predict y_t from X_t . The historical training pairs (X_τ, y_τ) exist only because training data is all in the past — both sides are knowable *to the modeller*, even though y_τ was not knowable *at time* τ .

Same setup as every LSTM in this project. The LSTMs take X_t (past features) and output $\hat{y}_t$ (predicted forward vol). HAR-RV is $\hat{y}_t = \text{OLS}(X_t)$ instead of $\hat{y}_t = \text{LSTM}(X_t)$ — same input-output contract, different function class. That's why HAR-RV's metrics are directly comparable to the LSTMs', and why the same Diebold–Mariano machinery applies to pairwise comparisons with A / H / B / O.

Scope notes

- No hyperparameter tuning. HAR-RV is OLS on three features — fit in milliseconds on $\sim 3,000$ train rows.
- No log transform of the target. Our LSTMs train in log-vol space for gradient reasons; OLS has no such need and `realized_vol_21d` is already non-negative and well-scaled.
- No GARCH(1,1) companion. Scoped to HAR-RV only per user decision.
- The `ignore/GARCH:Review.html` file supplied for the check was a paywalled ScienceDirect abstract only — no formulas or methodology — so it cannot verify target alignment independently.

Defensible paper claim

“We additionally compare against the HAR-RV benchmark of Corsi (2002/2004), adapted to our 21-day forward realized-volatility target: OLS regression of `realized_vol_21d(t)` on three backward-looking volatility components at 1-, 5-, and 21-day horizons. The adaptation preserves Corsi’s model structure while keeping the prediction target identical to that of our LSTM baselines, so HAR-RV’s MSE / MAE are directly comparable. The 1-day component uses $|r_t|$ (un-demeaned, as in Corsi); the 5- and 21-day components use the demeaned Andersen–Bollerslev form, matching the target’s own formula.”

Implementation pointer

`src/har_rv.py` — now implemented. CLI mirrors `src/ensemble.py`: writes `data/processed/test_predictions_harrrv.parquet` with columns `[har_rv, target, rv_d, rv_w, rv_m, naive]` indexed by date.

Results — live local run (2026-04-24)

HAR-RV fit in ~ 100 ms (OLS on 3 features $\times$ 3,690 train rows). No Colab training needed for this benchmark — HAR-RV is an econometric baseline, not a deep model.

Learned coefficients (match the prior: β_m dominates at $h = 21$):

Term	Value	Reading
intercept	$+2.72 \times 10^{-3}$	
β_d ($ r_t $, daily)	+0.046	tiny — daily shock carries little 21-day signal
β_w (5-day demeaned vol)	+0.234	moderate
β_m (21-day demeaned vol)	+0.494	dominant — best-matched predictor for forward 21-day target

Test metrics ($n = 1,461$; HAR-RV’s backward 21-day window drops fewer leading test rows than the LSTMs’ longer `seq_len`):

Predictor	MSE	RMSE	MAE	MAPE
naive (<code>rolling_std_21</code>)	2.23×10^{-5}	0.00472	0.00318	34.8%
HAR-RV	1.64×10^{-5}	0.00405	0.00282	31.5%

Diebold–Mariano vs naive ($h = 21$, Bartlett HAC, HLN-corrected):

Loss	DM	p	Verdict ($\alpha = 0.05$)
MSE	+1.76	0.079	tie (marginal at $\alpha = 0.10$)
MAE	+2.69	0.007	HAR-RV wins significantly

HAR-RV compared to our learned variants (informal)

Post-sentiment-merge nb 08 numbers used the LSTM common test index ($n = 1,434$); HAR-RV used $n = 1,461$. Numbers below are **not strictly row-for-row comparable** — proper alignment

+ DM pairwise HAR-RV vs each variant happens in task 6.

Predictor	MSE	MAE	MAPE	n
naive (A/H/B window)	2.22×10^{-5}	0.00313	34.6%	1,434
naive (HAR-RV window)	2.23×10^{-5}	0.00318	34.8%	1,461
HAR-RV	1.64×10^{-5}	0.00282	31.5%	1,461
Variant H baseline (<code>p_volatile</code>)	1.58×10^{-5}	0.00252	24.9%	1,434
Variant A ensemble	1.42×10^{-5}	0.00252	24.6%	1,434
Variant B ensemble (VIX)	1.40×10^{-5}	0.00251	26.9%	1,434

Interpretation (subject to row-aligned re-check in task 6):

- **HAR-RV closes most but not all of the gap** between naive and the deep variants. MSE: naive $2.23 \times 10^{-5} \rightarrow$ HAR-RV $1.64 \times 10^{-5} \rightarrow$ variant A 1.42×10^{-5} . HAR-RV captures $\sim 73\%$ of the naive-to-A MSE improvement using just a 3-feature OLS.
- HAR-RV $\approx$ Variant H on MSE (1.64×10^{-5} vs 1.58×10^{-5} — $\sim 4\%$ gap). Variant H is a single LSTM with `p_volatile` as a feature; HAR-RV matches it using OLS. If the DM row-aligned test returns a tie, that weakens the “regime signal as a feature adds unique value” argument for variant H.
- HAR-RV’s MAPE (31.5%) is noticeably worse than A/H/B ($\sim 25\%$). MAPE penalises relative errors on small-vol calm days — HAR-RV’s β_m -dominant prediction overshoots on calm days where 21-day backward vol is high but forward vol is low. The LSTMs handle this asymmetry better.
- **Variant A’s 13-17% MSE advantage over HAR-RV is real** in point estimate — the deep regime-split does add something over simple backward-vol weighting. Row-aligned DM tests will confirm whether this margin is statistically significant at $\alpha = 0.05$.

Defensible claims for the paper (pending row-aligned re-check)

- “The HAR-RV (Corsi 2002/2004) $h = 21$ benchmark significantly outperforms the naive 21-day persistence baseline (DM MAE $p = 0.007$), confirming that weighted multi-horizon past vol carries real signal beyond pure persistence.” $\leftarrow$ strongest claim, directly quotable.
- “The regime-split LSTM ensemble (variant A) beats HAR-RV on all three point metrics, with the largest gap on MAPE (24.6% vs 31.5%), suggesting the deep models handle the calm-day small-vol asymmetry better than an OLS on backward RV components.” $\leftarrow$ directional; conditional on DM confirmation.
- “HAR-RV is competitive with a single-LSTM regime-feature model (variant H) at $\sim 4\%$ MSE gap, suggesting the regime signal carries information that multi-horizon backward vol already captures.” $\leftarrow$ conditional; strong if DM returns a tie.

2026-04-24 (afternoon) — Variant O wiring: code complete, training pending

Why variant O exists

Variant O is the **pre-everything baseline** — trained without any of the additions we’ve made since Phase 6. It answers: *“did any of sentiment, VIX, or the HMM-as-feature architecture actually move the needle past where we started?”*

- **HMM input:** HMM_PRICE_VOLUME_FEATURES (11 feats — stationary price/volume only, no sentiment).
- **LSTM input:** LSTM_STATIONARY_FEATURES (5 feats — log_return, abs_return, oc_return, intraday_range, relative_volume_21d. No sentiment. No VIX. No p_volatile as a feature).
- **Architecture:** same 3-LSTM regime-split ensemble as variant A, but with regime labels from variant O’s own (sentiment-free) HMM. O and A differ *only* in feature sets, not in model structure.

Code changes to enable variant O

All backward-compatible — existing A / H / B invocations keep working unchanged. Variant O activates via new CLI args.

1. `src/train_hmm.py` (new module). Programmatic counterpart to `notebooks/03_hmm_regime.ipynb`. Parameterised on `--features-group {HMM_PRICE_VOLUME_FEATURES | HMM_FEATURES}` and `--output-suffix`. Saves `hmm_winner{suffix}.joblib`, `hmm_scaler{suffix}.joblib`, `hmm_meta{suffix}.joblib`, and `regime_probabilities{suffix}.parquet`. Delegates to functions already in `src/hmm_model.py` — no duplication.
2. `src/train_LSTM_regime.py`: added `--regime-probs-path` and `--hmm-meta-path` CLI args. Default `None` resolves to the current hardcoded paths (unchanged for A / B). Variant O passes `--regime-probs-path data/processed/regime_probabilities_0.parquet` and `--hmm-meta-path models/hmm_meta_0.joblib`. Also removed a dead `load_viterbi()` call whose output was immediately overwritten.
3. `src/ensemble.py`: same two CLI args, same default-preservation. Variant O’s invocation:
`python -m src.ensemble --variant-suffix _0 --regime-probs-path ... --hmm-meta-path ...`

Notebook integration

`notebooks/08_seven_model_comparison.ipynb` now has six new cells (9–14) placed after variant-B training and before “Produce test-set predictions”. They orchestrate the variant-O pipeline end-to-end in four stages: (1) retrain HMM, (2) train baseline LSTM, (3) train regime LSTMs with variant-O regime labels, (4) run ensemble with variant-O soft probabilities.

Exact CLI invocations (see nb 08 cells 11–14 for the shell-magic variants):

```
python -m src.train_hmm \
    --features-group HMM_PRICE_VOLUME_FEATURES \
    --output-suffix _0
```

```
python -m src.train_LSTM_baseline \
```

```

--features <LSTM_STATIONARY_FEATURES> \
--output-prefix lstm_baseline_0

python -m src.train_LSTM_regime \
--features <LSTM_STATIONARY_FEATURES> \
--regime both \
--output-suffix _0 \
--regime-probs-path data/processed/regime_probabilities_0.parquet \
--hmm-meta-path      models/hmm_meta_0.joblib

python -m src.ensemble \
--variant-suffix _0 \
--regime-probs-path data/processed/regime_probabilities_0.parquet \
--hmm-meta-path      models/hmm_meta_0.joblib

```

Artifacts variant O will produce on first run

- models/hmm_winner_0.joblib, models/hmm_scaler_0.joblib, models/hmm_meta_0.joblib
- data/processed/regime_probabilities_0.parquet
- models/lstm_baseline_0.pt (+ scaler), models/lstm_calm_0.pt (+ scaler), models/lstm_volatile_0.pt (+ scaler)
- data/processed/test_predictions_0.parquet
(columns: baseline, calm, volatile, target, p_calm, p_volatile, ensemble)

Status

Code complete, not yet trained. Training will run on Colab once the remaining comparison / benchmark pieces (HAR-RV module + 5-way nb-08 table cells) are in place. No training artifacts exist locally yet — running nb 08 cells 11–14 will produce them.

Expected role in the 5-way comparison table

Variant O is the **floor** row in the final comparison table {naive, A, H, B, O, HAR-RV}. Key framing questions:

- Does variant-A **ensemble with sentiment** beat variant O without sentiment? → quantifies incremental value of AAI bullish/bearish columns.
- Does variant H (HMM-as-feature) beat variant O? → tests whether exposing the regime signal to the LSTM via any route helps over no-regime-signal at all.
- Does variant B (VIX family) beat variant O? → isolates the value of forward-looking implied-vol features over price-only.

If variant O ties with A / H / B on DM tests, that’s additional evidence for the structural-limits framing: **none of the post-phase-6 feature expansions mattered at this horizon**, consistent with F1 being a data-rate $\times$ horizon property rather than a feature-set problem. If variant O loses significantly, that’s evidence the feature additions do carry real signal and we should reframe accordingly.